

Tugas 2 Kalkulus

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1.) Diketahui : $y = 3x^2 - 2x + 5$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0}$$

$$\frac{3(x+h)^2 - 2(x+h) + 5 - (3x^2 - 2x + 5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h} = \lim_{h \rightarrow 0} (6x + 3h - 2) = 6x - 2$$

2.) Diketahui $f(x) = x^2 - 1$

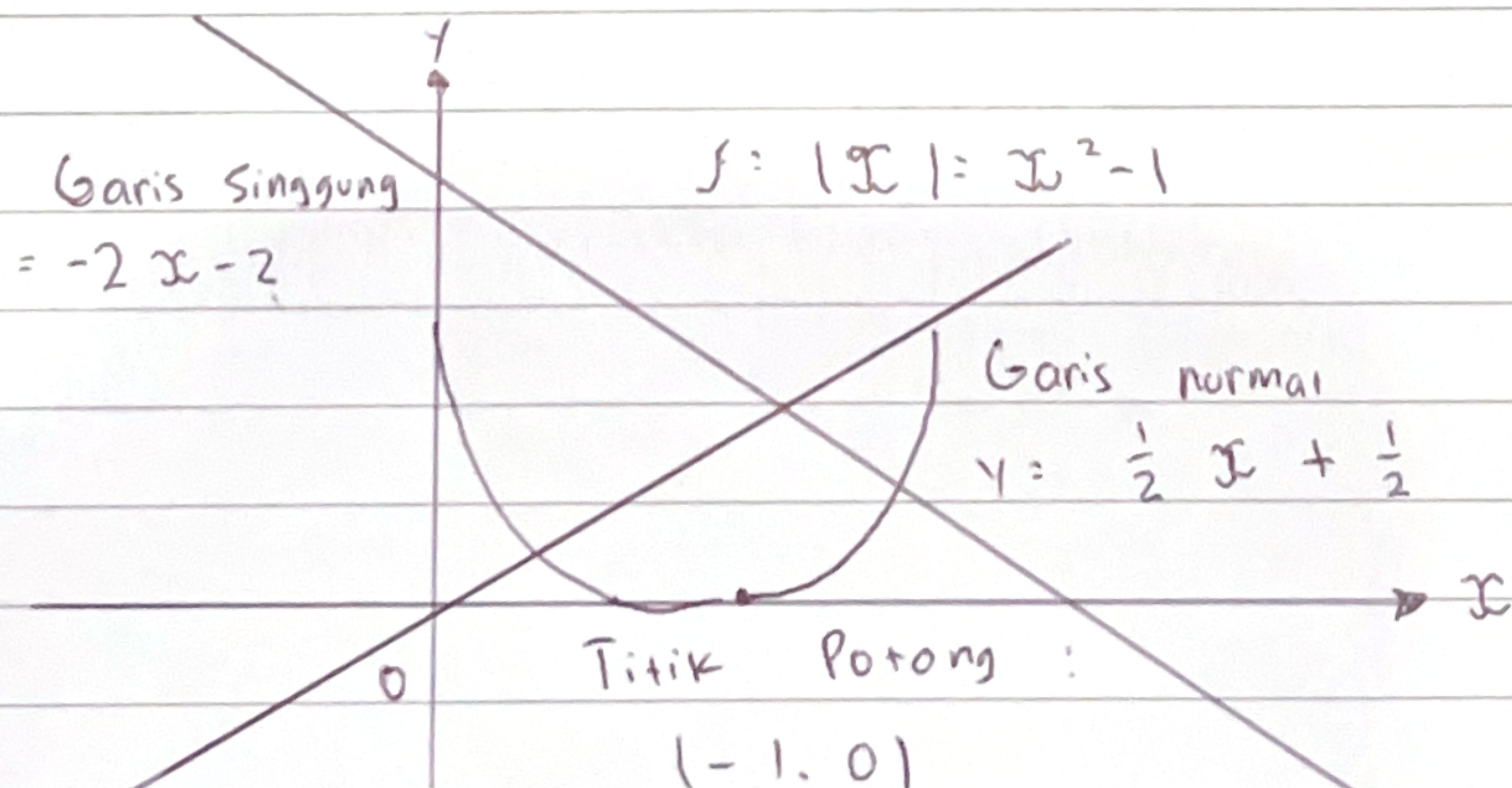
a. Persamaan garis Singgung di $x = 1$:

$$f'(1) = 2x = 2 \quad f(1) = 0 \Rightarrow y - 0 = 2(x - 1) \Rightarrow y = 2x - 2$$

b. garis normal ; $= -\frac{1}{2} = \frac{1}{2}$

$$y = \frac{1}{2}(x + 1) \Rightarrow y = \frac{1}{2}x + \frac{1}{2}$$

c.



3.) Diketahui : $y = x \cos (2x)$:

a. $y'''(x) = 8x \sin (2x) - 12 \cos (2x)$

b. $y'''(\pi) = 8\pi \sin (2\pi) - 12 \cos (2\pi) = 0 - 12(1) = -12$

4.) a. Jika $y = \ln |\sin (x^3 - 1)|$. maka :

$$\frac{dy}{dx} = \frac{3x^2}{\tan (x^3 - 1)}$$

b. $F(xy) = 2x^2y - \sin (xy^2) + e^{-xy} = 10$.

$$\frac{dy}{dx} = \frac{y(1 - 4xe^{xy} + ye^{xy} \cos (xy^2) + 1)}{x(2xe^{xy} - 2ye^{xy} \cos (xy^2) - 1)}$$

5.) Diketahui : $f(x) = \sin (2x)$

a. $f(x) \approx -2x + \frac{4}{3} \left(x - \frac{\pi}{2}\right)^3 + \pi$

b. $f(x) \approx 2x - \frac{4}{3} x^3$